Part II Report

Comparative Analysis of GO-Enriched Communities in Bacterial STRING Networks

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1. Introduction and Analytical Rationale

This report extends our Part I analysis of bacterial protein interaction networks from within-species community detection to a comparative interpretation of functional organisation across species. Rather than relying solely on exact protein overlap, we represent each detected module by its enriched Gene Ontology (GO) terms. These descriptors capture biological processes, molecular functions and cellular components associated with the proteins in a module. By comparing modules at the functional level, we can identify functionally conserved modules even when their exact protein composition differs between bacterial species.

The analysis highlights both conserved and species-specific module-level functions and provides an interpretive framework for understanding how related processes are organised across diverse bacterial systems.

2. Data Sources and Network Preparation

We analysed four bacterial protein-protein interaction networks derived from the STRING database. Each dataset comprised an edge file containing weighted functional or physical associations and an annotation file mapping proteins to GO terms. Nodes in the network represent proteins, and edges are weighted by STRING combined scores. For each organism we constructed an undirected graph, filtered edges by a confidence threshold to remove low-confidence interactions and used proteins with GO annotations for downstream functional enrichment analysis. The datasets included two virulence-related networks and two resistance-related networks: B5 (Salmonella virulence), B6 (Vibrio cholerae virulence), B21 (Bacillus subtilis resistance) and B22 (Erwinia resistance).

3. Module Representation and Comparative Framework

Modules were detected in each network using the Louvain community detection algorithm. Functional characterization of modules was performed via GO enrichment analysis using a hypergeometric test and Benjamini-Hochberg correction (FDR ≤ 0.05). Each module was represented by its set of significantly enriched GO terms. To compare modules across and within species we computed pairwise similarities between their term

sets. After experimenting with several set-based measures (e.g. Jaccard, Dice), we selected the overlap coefficient as the final similarity metric because it is sensitive to inclusion relationships between small and large term sets. The similarity threshold was tuned separately for each comparison based on the distribution of observed overlap scores: for the B21/B22 resistance pair a threshold of 0.30 was applied, and for the B5/B6 virulence pair a threshold of 0.05 was used to accommodate generally lower overlap values.

4. Cross-Species Comparative Analysis

Cross-species comparisons were performed to identify modules that appear to represent conserved biological functions across different bacterial species. We compared two network pairs, where each pair represented a similar biological theme.
In the virulence pair (B5 vs B6), modules representing secretion processes, motility, biofilm formation and host interactions were compared between Salmonella and Vibrio cholerae. In the resistance pair (B21 vs B22), modules associated with transport, antibiotic response, cell-wall biosynthesis and membrane organisation were compared between Bacillus subtilis and Erwinia. A module from one species was considered conserved if at least one module in the other species surpassed the similarity threshold. Conserved modules were interpreted biologically, and differences were discussed in relation to species-specific adaptations, annotation completeness and network structure.

5. Within-Species Comparative Analysis

Within-species comparisons examined whether multiple modules in the same network shared sufficiently similar enrichment profiles to suggest functional overlap or sub-networks of broader biological processes. In the virulence networks (B5 and B6), no module pairs exceeded the similarity threshold, suggesting that secretion, motility, biofilm formation, and host interaction functions were represented as relatively distinct modules in the detected network partition. In the resistance networks, B21 showed one internal match between module B21_M3 (gene expression) and B21_M4 (response to antibiotic/transport), suggesting partial overlap between transcriptional regulation and stress response modules. No internal matches were observed in B22.

6. Results

The main findings of the comparative analysis are summarised below integrating quantitative similarity measures with biological interpretation**.** Results are presented separately for within-species and cross-species comparisons, accompanied by concise summaries of module counts and conserved pairs. Quantitative evidence is linked to biological interpretation to provide a coherent analytical narrative.

6.1 Required Pairwise Identification of Conserved Modules

6.1.1 Conserved Modules in the Virulence Interactomes of Salmonella (B5) and Vibrio cholerae (B6)

Modules detected in the virulence networks were compared using the **overlap coefficient**. Because the overall overlap values were modest, a permissive similarity threshold of **0.05** was used to identify conserved modules. In this B5/B6 comparison, three module pairs surpassed the threshold. The strongest match was between **B5_M1** and **B6_M2**: all GO terms enriched in the secretion module of *Salmonella* were a subset of the larger secretion module in *Vibrio cholerae*, yielding an overlap coefficient of 1.0. This indicates a highly conserved secretion function across the two pathogens.
[Korotkov, K. V., Sandkvist, M., & Hol, W. G. J. (2012)]
A second match involved **B5_M2** and **B6_M4**, both enriched for extracellular or interaction-related terms. Although the overlap score was high, the *Salmonella* module contained few terms and the biological interpretation remains tentative. The third pair, **B5_M3** and **B6_M6**, showed only partial similarity with an overlap coefficient of 0.11 and was classified as a weak or partial match. No other modules exceeded the 0.05 threshold, suggesting that many virulence modules are species-specific. Overall, the virulence comparison highlighted conservation of secretion machinery while suggesting divergence in other virulence-related processes.

6.1.2 Conserved Modules in the Resistance Interactomes of Bacillus subtilis (B21) and Erwinia (B22)

For the resistance networks, we compared modules using the overlap coefficient and applied a stricter similarity threshold of **0.30** to focus on robust functional overlaps. Under this criterion the B21/B22 comparison yielded several conserved module pairs. The most significant match linked **B21_M7** and **B22_M6**, both enriched for cell-wall and peptidoglycan biosynthesis terms, highlighting conservation of cell-wall remodelling mechanisms and antibiotic binding functions.
[Egan, A. J. F., Errington, J., & Vollmer, W. (2020)]
Another strong match connected **B21_M2** to **B22_M3**, revealing a conserved transport/localisation module. Additional matches clustered modules **B21_M4**, **B21_M6** and **B22_M1**, which collectively captured response to antibiotic, membrane transport and cell envelope biogenesis. Modules without significant overlap above the 0.30 threshold were considered species-specific. Together, these matches indicate that resistance networks share conserved modules related to cell-wall maintenance, transport and antibiotic response while also exhibiting species-specific expansions and specialisations.

6.2 Comparative Functional Interpretation of Detected Modules

The enriched GO terms associated with each module reveal recurring biological themes across and within species. In the virulence networks, modules correspond to secretion systems, flagellum-mediated motility, biofilm formation, host interaction and enzymatic activities such as penicillin binding. These themes align with known virulence mechanisms in Gram-negative pathogens. In the resistance networks, modules

represent transcriptional regulation, gene expression, membrane transport, antibiotic response, biosynthesis of antibiotics, carboxypeptidase activity and peptidoglycan metabolism. Comparing these themes across species shows that secretion and cell-wall modules are highly conserved, whereas motility and biofilm formation appear more species-specific. Within species, the absence of significant overlaps in B5, B6 and B22 suggests clear functional partitioning, while the partial overlap in B21 hints at interplay between gene expression and stress-response programmes.

7. Discussion and Analytical Limitations:

The comparative analysis supports the view that bacterial modules may be biologically conserved even when they are not structurally identical. In this study, module comparison was performed at the level of enriched GO-term profiles rather than exact protein membership, following the project rationale that related biological roles may be implemented by partially different protein sets across species. This distinction between structural similarity and functional similarity is important for interpreting the results: two modules may differ substantially in their exact protein composition and still represent the same broad biological function if their enriched descriptors overlap in a meaningful way. This was particularly evident in the cross-species comparisons, where the strongest conserved pairs reflected shared virulence- or resistance-related functions rather than exact molecular equivalence.

The Part II experiments also showed that the interpretation of conservation depends strongly on the chosen similarity definition. Several similarity metrics were explicitly tested on the same all-against-all module comparisons: jaccard_similarity, tweaked_jaccard_similarity, overlap_coefficient, and dice_similarity. In the B21/B22 comparison, using the shared threshold of 0.30, Jaccard identified only 1 conserved pair, tweaked Jaccard identified 2 conserved pairs, Dice identified 2 conserved pairs, whereas overlap coefficient identified 6 conserved pairs in total, including 5 cross-species and 1 within-species pair. In the B5/B6 comparison, using the shared threshold of 0.05, Jaccard identified only 1 conserved pair, Dice also identified 1, tweaked Jaccard identified 5 conserved pairs, including within-species matches, whereas overlap coefficient identified 3 cross-species conserved pairs and no within-species conserved pairs. These experiments indicate that the overlap coefficient was more sensitive to biologically meaningful containment relationships between modules of unequal functional-set sizes, while Jaccard and Dice were more conservative and tended to suppress partial but still interpretable overlaps. At the same time, the tweaked Jaccard measure appeared more permissive and, in the B5/B6 case, produced extra within-species matches that were not retained in the final analysis. For this reason, overlap coefficient was selected as the final similarity metric.

Threshold sensitivity analysis further demonstrated that conservation results are not fixed properties of the data, but depend on the interpretive cutoffs applied to the similarity scores. For B21/B22, increasing the threshold from 0.05 to 0.70 reduced the number of conserved cross-species pairs from 11 to 1, and reduced the number of

within-species conserved pairs in B21 from 5 to 0, while B22 showed no within-species conserved pairs across all tested thresholds. For B5/B6, the final threshold was set lower than for B21/B22 because the virulence comparison produced weaker overall overlap values, and a stricter cutoff would have eliminated biologically relevant cross-species correspondences. These experiments justify the use of pair-specific thresholds rather than a single universal cutoff for all datasets. They also show that the boundary between “conserved”, “partially overlapping”, and “species-specific” modules is partly methodological and not purely biological.

An additional Part II experiment examined different methods for clustering the module-similarity graph in order to define higher-level conserved clusters. For B5/B6, connected components, greedy modularity, label propagation, and Louvain all produced the same summary outcome: 3 conserved clusters, each spanning both bacteria, with the same average internal similarity and the same modularity score. This indicates that, once the similarity threshold was applied, the B5/B6 similarity graph was simple and robust enough that the clustering method had little practical effect on the resulting higher-level structure. In contrast, for B21/B22 the final implementation retained Louvain as the conserved-cluster method, consistent with the general preference throughout the project for graph-native community detection approaches. These observations strengthen the interpretation that some conserved patterns are highly stable across methodological choices, whereas weaker patterns depend more strongly on the chosen comparison framework.

From a biological perspective, the results suggest that the strongest conserved modules correspond to core functional systems that remain important across species, even if the detailed module composition differs. In the virulence pair, the main conserved signal centered on secretion-related biology, whereas in the resistance pair the most convincing conservation was associated with cell wall / peptidoglycan processes, transport, and antibiotic-related response. By contrast, modules related to motility, biofilm formation, toxin-associated functions, or broader regulatory processes appeared more variable and were therefore interpreted as more species-specific or context-dependent. This supports the broader biological idea that essential or recurrent functional programs may be preserved at the level of role or process, while their precise molecular realization can diverge across bacterial lineages.

Several uncertainties must nevertheless be acknowledged. First, all comparative conclusions depend on the upstream module detection step, including the edge-weight cutoff, the selected community-detection algorithm, and the minimum module size. Changes in these parameters can alter module boundaries and therefore alter the enrichment profiles that are later compared. Second, the analysis depends on the completeness and consistency of functional annotations. GO-term coverage is not necessarily uniform across species, and sparse or uneven annotation can reduce observed similarity even when the underlying biology is related. Third, the comparison strategy represents each module only by its significant enriched terms and ignores other

potentially informative properties, such as detailed network topology, hub structure, or explicit homology relationships. Finally, all set-based similarity measures simplify biological function into discrete term overlap and do not fully capture the hierarchical relationships among GO terms or the different biological weights of different annotations.

Overall, the analytical framework is valuable because it makes comparative network interpretation reproducible, biologically interpretable, and less dependent on exact protein overlap alone. At the same time, the Part II experiments show clearly that conservation calls are sensitive to metric choice, threshold choice, and clustering strategy. The most defensible conclusion is therefore not that all detected matches are equally strong, but that the strongest patterns are robust enough to support biological interpretation, whereas weaker matches should be treated as candidates for partial conservation rather than definitive evidence of shared module identity.

Finally, it is important to note that the observed conservation patterns are consistent with the biological context of the input datasets. The compared bacterial pairs (B5/B6 and B21/B22) represent functionally related systems, specifically virulence-associated and resistance-associated networks, and therefore a certain degree of functional conservation is expected. The detection of conserved modules in key biological processes such as secretion systems, cell wall biosynthesis, transport and antibiotic response supports this expectation and reinforces the biological plausibility of the results.

8. Proposed Creative Extension: Cross-Species Module Family Discovery

The creative component of this project extends the comparative analysis beyond isolated pairwise comparisons (B5/B6 and B21/B22) toward a unified framework that analyses modules across multiple bacterial species.

	ID	taxid	Organism	Group	Network type	Role in creative workflow	Evidence
0	B22	None	Erwinia	Resistance	Resistance network	Target bacterium	Manual mapping for the fixed report IDs
1	B1	665029	Erwinia amylovora	Virulence screening	Virulence screening network	Screening bacterium	STRING interaction file taxid prefix
2	B10	224308	Bacillus subtilis	Virulence screening	Virulence screening network	Screening bacterium	STRING interaction file taxid prefix
3	B11	395492	Rhizobium leguminosarum	Virulence screening	Virulence screening network	Screening bacterium	STRING interaction file taxid prefix
4	B2	267608	Rhodobacter sphaeroides	Virulence screening	Virulence screening network	Screening bacterium	STRING interaction file taxid prefix
5	B3	56449	Xanthomonas campestris	Virulence screening	Virulence screening network	Screening bacterium	STRING interaction file taxid prefix
6	B4	272561	Chlamydia trachomatis	Virulence screening	Virulence screening network	Screening bacterium	STRING interaction file taxid prefix
7	B5	59204	Salmonella enterica	Virulence	Virulence network	Screening bacterium	STRING interaction file taxid prefix
8	B6	243277	Vibrio cholerae	Virulence	Virulence network	Screening bacterium	STRING interaction file taxid prefix
9	B7	1314	Streptococcus pneumoniae	Virulence screening	Virulence screening network	Screening bacterium	STRING interaction file taxid prefix
10	B8	93061	Staphylococcus aureus	Virulence screening	Virulence screening network	Screening bacterium	STRING interaction file taxid prefix
11	B9	223283	Pseudomonas syringae	Virulence screening	Virulence screening network	Screening bacterium	STRING interaction file taxid prefix

Instead of focusing only on species-to-species comparisons, this extension treats each detected module as an independent analytical unit. Each module is represented by its enriched GO term profile together with structural properties such as size and density, enabling a higher-level representation in which modules themselves become the objects of analysis.

The proposed extension aims to identify module families, defined as groups of modules across multiple species that share similar functional signatures. This allows classification of modules into:
• Conserved module families (e.g., cell wall or secretion systems appearing across species),
• Species-specific module families (e.g., biofilm or motility modules limited to certain organisms),
• Functionally rewired modules, where similar biological roles are implemented using different network structures.

This module-level perspective enables identification of higher-order biological patterns that are not observable in simple pairwise analysis.

9. Methodological Plan for the Proposed Creative Extension

The proposed extension follows a two-layer analytical framework that generalizes the pairwise comparison approach into a multi-species setting.

First layer: protein–protein interaction networks from multiple bacterial species are analysed independently. For each species, modules are detected using the same pipeline applied in Part I, including network construction, Louvain community detection, and GO-term enrichment analysis. This ensures methodological consistency across datasets.
Second layer: modules derived from all species are compared directly. A module similarity network is constructed, where each node represents a module and edges connect modules with similar enriched GO term profiles. Similarity is defined using the overlap coefficient, as in Part II.

This extension was implemented as a configurable screening pipeline. The user defines the target bacterium, input folders, similarity metric, similarity threshold, and the minimum number of supporting datasets required for a module to be considered conserved.

In the example run, B22 was used as the target bacterium and screened against virulence datasets using overlap_coefficient with a threshold of 0.3 and a requirement of support from at least two independent datasets.
B22 was selected as the target bacterium not only because Part II had already identified strong conserved modules for it in comparison with B21, but also because it provided a strong test case for the creative workflow. In the earlier analysis, B22 contained clear functional modules related to cell wall biology, peptidoglycan metabolism, penicillin-binding functions, and transport/localization. These functions are not limited to a single bacterial lifestyle and are more likely to represent broadly conserved bacterial systems.

This made B22 a suitable target for screening against a wider directory of virulence-related datasets. Unlike a virulence target, which would be expected to match mostly phenotype-specific processes such as secretion, motility, or biofilm formation, a resistance target such as B22 allows testing whether more general bacterial core functions can still be recovered across a more heterogeneous comparison set. In other words, the creative workflow was designed to ask whether modules first observed in a resistance context would also receive repeated support from multiple virulence-network inputs.

In addition, B22 included small but functionally distinct modules, which made their enriched signatures easier to interpret and compare. This was especially useful in the creative pipeline, where target modules were retained only if they were supported by matches from at least two distinct screening inputs. Therefore, choosing B22 strengthened both the biological interpretation and the methodological robustness of the directory-based screening analysis.

Under these settings, four modules from B22 passed the filtering criteria: B22_M1, B22_M3, B22_M4, and B22_M6.

Two of Erwinia's modules: B22_M3 and B22_M6 consistently match modules from multiple virulence organisms. B22_M3 aligns with localization-enriched modules in *Erwinia amylovora*, *Bacillus subtilis*, *Rhizobium leguminosarum*, *Chlamydia trachomatis*, *Vibrio cholerae*, *Streptococcus pneumoniae* and *Pseudomonas syringae*. B22_M6 shows strong overlap with proteolysis-oriented modules in *Bacillus subtilis*, *Chlamydia trachomatis*, *Vibrio cholerae*, *Streptococcus pneumoniae* and *Pseudomonas syringae*. The highest similarity scores (≥0.93) occur when B22_M6 matches proteolysis modules in *Bacillus subtilis* and *Vibrio cholerae* and when B22_M3 matches localization modules in *Rhizobium leguminosarum*.

Localization is the dominant shared signal**.** Most matches are driven by GO terms related to protein/peptide localization. This suggests that Erwinia's B22_M3 module, which is itself enriched for transport and localization functions, has a conserved role across many pathogens. Proteolysis is the second most common signal, associated primarily with B22_M6.

Additional, less frequent signals. Two single-match signals appear: lipooligosaccharide biosynthetic process (B22_M1 matching the LPS biosynthesis module B10_M9 in *Bacillus subtilis*) and cellular response to stimulus (B22_M4 matching B8_M3 in *Staphylococcus aureus*). These matches have lower similarity scores (≈0.38–0.44), indicating weaker functional overlap.

The fact that B22_M3 and B22_M6 match modules from organisms as diverse as *Bacillus subtilis* (a Gram-positive soil bacterium), *Vibrio cholerae* (a Gram-negative enteric pathogen) and *Pseudomonas syringae* (a plant pathogen) suggests that the

underlying processes ,protein localization/transport and proteolysis, are fundamental and conserved across bacterial clades(groups of organisms that share a common ancestor and include all of its descendants.).

B22_M1 (lipooligosaccharide/stress-response module) this module overlapped with Bacillus subtilis module B10_M9 through a shared “lipooligosaccharide biosynthetic process” signal, and with B10_M2 through a general “cellular response to stimulus” signal. It also matched Staphylococcus aureus module B8_M5 in the lipooligosaccharide biosynthesis category, but with an even lower similarity.

These lower scores suggest that B22_M1 encodes functions (LPS biosynthesis and stress response) that are only modestly similar to modules in a few virulence networks.

B22_M4 (stress-response module) matched B8_M3 in Staphylococcus aureus via the “cellular response to stimulus” signal, with a similarity score of 0.40. In other words, B22_M4 corresponds to a stress-response module that is weakly conserved and does not recur across the broader screening set.

Modules that appeared only in isolated comparisons without repeated support were filtered out. This highlights a key advantage of the approach: conserved candidates are supported by consistent evidence rather than single pairwise matches.

Overall, this extension provides a more robust framework for identifying conserved functional systems across multiple datasets, reducing the influence of weak or dataset-specific similarities.

	Comparison	Compared against ID	Compared against organism	Target module	Compared module	Shared functional signal	Similarity score
0	Erwinia vs Erwinia amylovora	B1	Erwinia amylovora	B22_M3	B1_M4	Localization	0.666667
1	Erwinia vs Bacillus subtilis	B10	Bacillus subtilis	B22_M3	B10_M5	Localization	1.000000
2	Erwinia vs Bacillus subtilis	B10	Bacillus subtilis	B22_M6	B10_M11	Proteolysis	0.931818
3	Erwinia vs Bacillus subtilis	B10	Bacillus subtilis	B22_M3	B10_M2	Localization	0.666667
4	Erwinia vs Bacillus subtilis	B10	Bacillus subtilis	B22_M1	B10_M9	Lipooligosaccharide biosynthetic process	0.437500
5	Erwinia vs Bacillus subtilis	B10	Bacillus subtilis	B22_M4	B10_M2	Cellular response to stimulus	0.400000
6	Erwinia vs Rhizobium leguminosarum	B11	Rhizobium leguminosarum	B22_M3	B11_M6	Localization	1.000000
7	Erwinia vs Chlamydia trachomatis	B4	Chlamydia trachomatis	B22_M3	B4_M9	Localization	0.500000
8	Erwinia vs Chlamydia trachomatis	B4	Chlamydia trachomatis	B22_M6	B4_M2	Proteolysis	0.390244
9	Erwinia vs Vibrio cholerae	B6	Vibrio cholerae	B22_M6	B6_M6	Proteolysis	1.000000
10	Erwinia vs Vibrio cholerae	B6	Vibrio cholerae	B22_M3	B6_M2	Localization	0.666667
11	Erwinia vs Streptococcus pneumoniae	B7	Streptococcus pneumoniae	B22_M3	B7_M3	Localization	0.500000
12	Erwinia vs Streptococcus pneumoniae	B7	Streptococcus pneumoniae	B22_M6	B7_M2	Proteolysis	0.454545
13	Erwinia vs Streptococcus pneumoniae	B7	Streptococcus pneumoniae	B22_M3	B7_M6	Localization	0.333333
14	Erwinia vs Staphylococcus aureus	B8	Staphylococcus aureus	B22_M4	B8_M3	Cellular response to stimulus	0.400000
15	Erwinia vs Staphylococcus aureus	B8	Staphylococcus aureus	B22_M1	B8_M5	Lipooligosaccharide biosynthetic process	0.384615
16	Erwinia vs Pseudomonas syringae	B9	Pseudomonas syringae	B22_M6	B9_M2	Proteolysis	0.954545
17	Erwinia vs Pseudomonas syringae	B9	Pseudomonas syringae	B22_M3	B9_M4	Localization	0.666667

10. Conclusion

Part II extended the project from within-network community detection to a comparative analysis of bacterial functional organisation. By representing modules using their enriched GO term profiles and comparing them via the overlap coefficient, we were able to identify both conserved modules and species-specific patterns across bacterial interaction networks. In the virulence networks, the secretion module emerged as a strongly conserved element, whereas other modules diverged, reflecting distinct strategies for host interaction and nutrient acquisition. In the resistance networks, modules involved in cell wall/peptidoglycan biosynthesis, transport and antibiotic response were found to be conserved across species. These findings demonstrate that functional conservation can occur even when the underlying protein compositions differ and underscore the importance of calibrating similarity thresholds to capture meaningful overlaps without introducing noise.

The creative extension introduced a higher-level framework in which modules themselves become the objects of comparison. By constructing a module similarity network and identifying module families supported by multiple external datasets, this extension allows detection of conserved functional systems that recur across several bacterial species, as well as identification of species-specific or functionally rewired modules. Together, the comparative analysis and creative extension provide a coherent pathway from network-level community detection to cross-species biological interpretation, highlighting secretion and cell wall modules as core conserved elements of bacterial biology, while motility, biofilm and regulatory modules exhibit greater diversity.

11. Submission Components

Part I

File Type	Filename
Part I Report	`mini_project_PartI_Report_Final.docx`
Part I Code	`project_sumbit.ipynb`
Module Summary Tables	`run_<timestamp>/B5_module_summary.csv, B6_module_summary.csv, B21_module_summary.csv, B22_module_summary.csv`
Enrichment Detailed Tables	`run_<timestamp>/B5_module_terms_detailed.csv, B6_module_terms_detailed.csv, B21_module_terms_detailed.csv, B22_module_terms_detailed.csv`
Module Enrichment Files	`run_<timestamp>/B*_M*_enrichment.csv`
Network Figures	`run_<timestamp>/B*_M*_network.png`

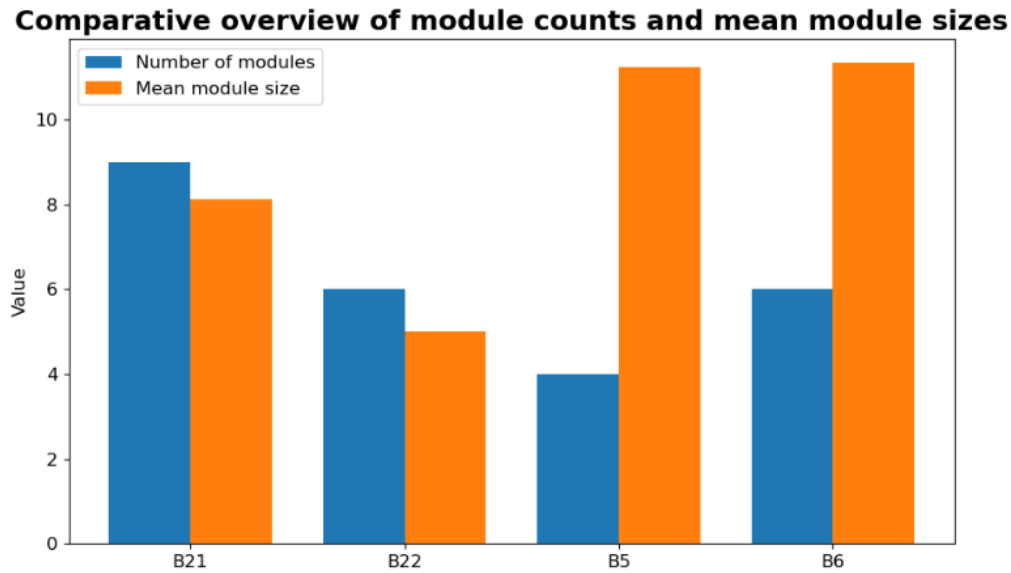
Part II

File Type	Filename
Part II Report	`PartII_Report_Final_filled.docx`
Part II Code	`project_sumbit.ipynb`
Cross-Species Comparison Tables	`run_<timestamp>/B5_B6_overlap_module_comparison.csv, B21_B22_overlap_module_comparison.csv`
Conserved Modules Tables	`run_<timestamp>/B5_B6_overlap_conserved_modules.csv, B21_B22_overlap_conserved_modules.csv`
Similarity Graphs (Cross-Species)	`run_<timestamp>/*_cross_species_similarity_graph.png`
Similarity Graphs (Within-Species)	`run_<timestamp>/*_within_species_similarity_graph.png`
Additional Visualizations	`run_<timestamp>/heatmaps and supplementary plots`

Creative Extension

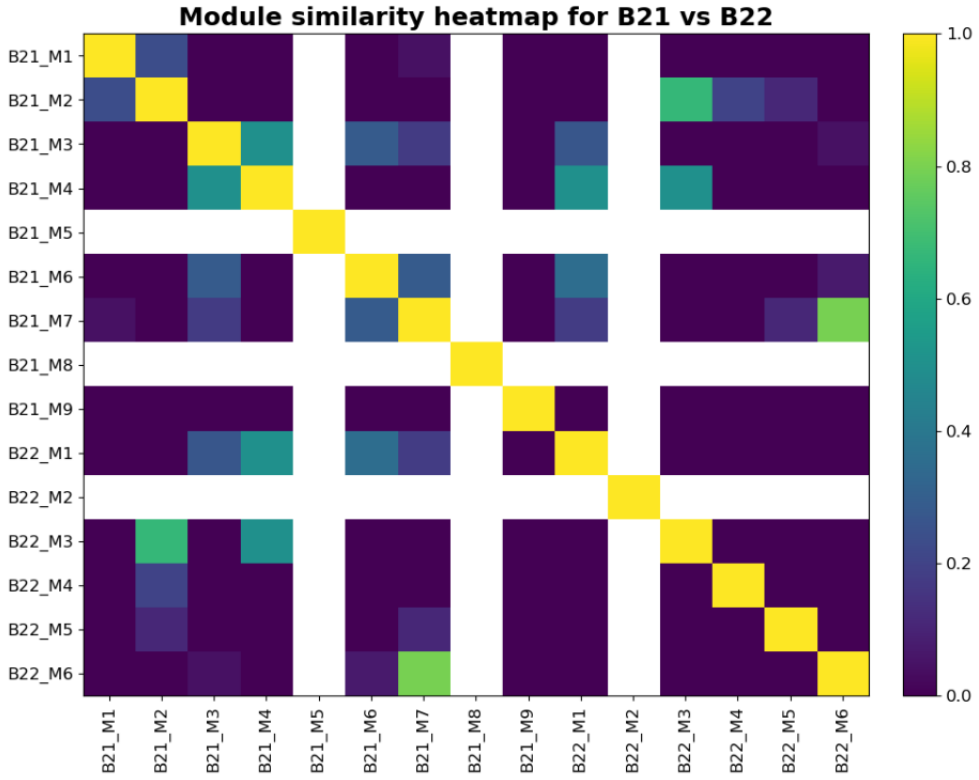
File Type	Filename
Creative Extension Code	`project_sumbit.ipynb (Part II section)`
Creative Extension Results	`run_<timestamp>/ (generated via extended screening pipeline)`
Creative Extension Tables & Graphs	`Displayed within notebook output (no separate export files)`

Figure 1. Comparative overview of module counts and sizes across bacterial networks:

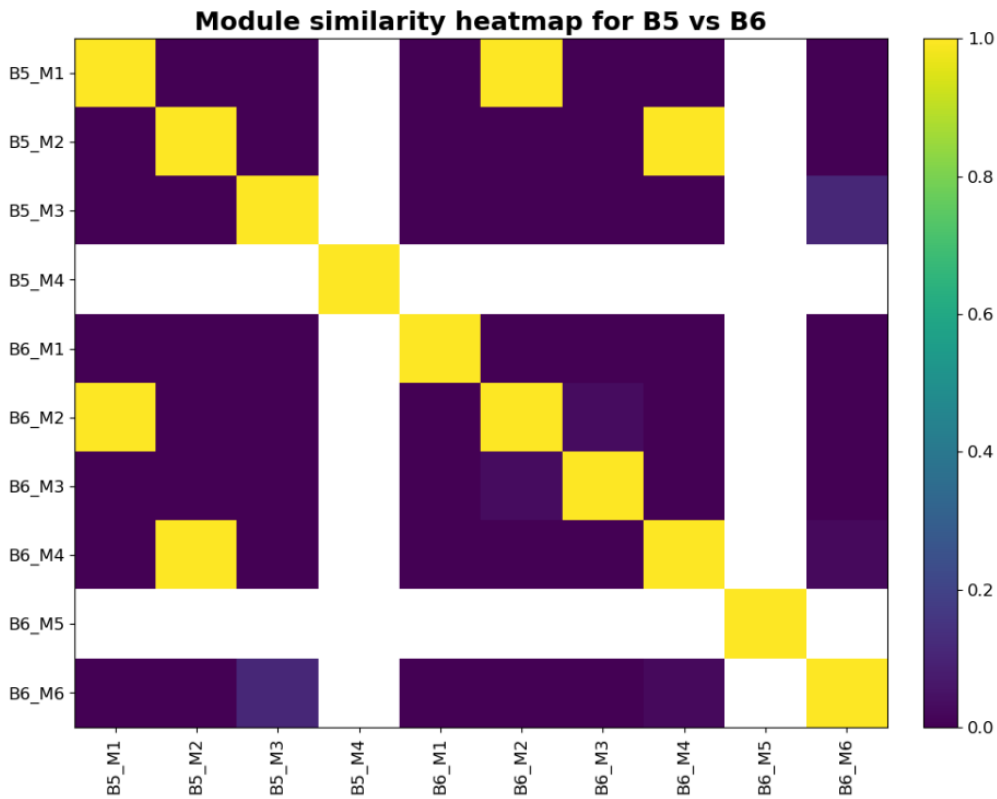


	Bacterium	Number of modules	Mean module size
0	B21	9	8.111111
1	B22	6	5.000000
2	B5	4	11.250000
3	B6	6	11.333333

Figure 2. Heatmap of module similarity scores based on enriched biological descriptors:



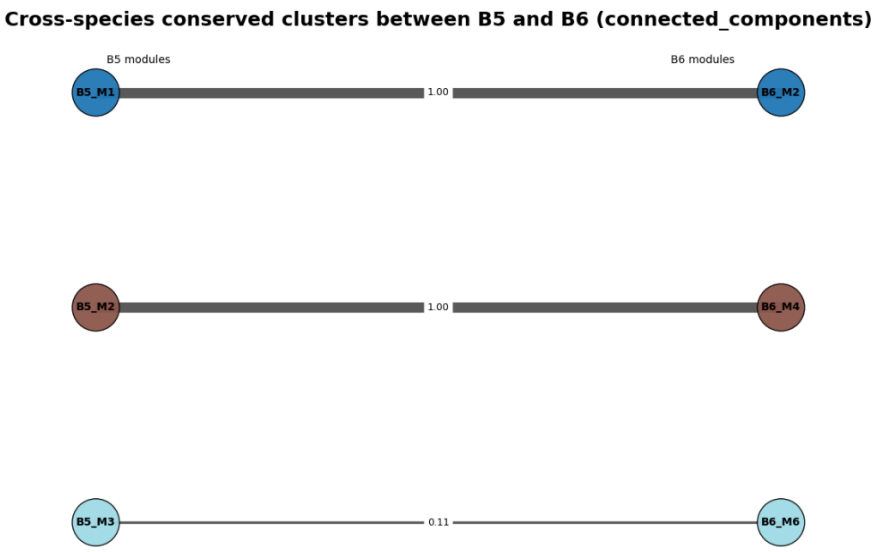
	B21_M1	B21_M2	B21_M3	B21_M4	B21_M5	B21_M6	B21_M7	B21_M8	B21_M9	B22_M1	B22_M2	B22_M3	B22_M4	B22_M5	B22_M6
B21_M1	1.000000	0.235294	0.000000	0.0	NaN	0.000000	0.044444	NaN	0.0	0.000000	NaN	0.000000	0.0	0.000000	0.000000
B21_M2	0.235294	1.000000	0.000000	0.0	NaN	0.000000	0.000000	NaN	0.0	0.000000	NaN	0.666667	0.2	0.111111	0.000000
B21_M3	0.000000	0.000000	1.000000	0.5	NaN	0.285714	0.173077	NaN	0.0	0.269231	NaN	0.000000	0.0	0.000000	0.045455
B21_M4	0.000000	0.000000	0.500000	1.0	NaN	0.000000	0.000000	NaN	0.0	0.500000	NaN	0.500000	0.0	0.000000	0.000000
B21_M5	NaN	NaN	NaN	NaN	1.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
B21_M6	0.000000	0.000000	0.285714	0.0	NaN	1.000000	0.285714	NaN	0.0	0.357143	NaN	0.000000	0.0	0.000000	0.071429
B21_M7	0.044444	0.000000	0.173077	0.0	NaN	0.285714	1.000000	NaN	0.0	0.178571	NaN	0.000000	0.0	0.111111	0.795455
B21_M8	NaN	NaN	NaN	NaN	NaN	NaN	NaN	1.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN
B21_M9	0.000000	0.000000	0.000000	0.0	NaN	0.000000	0.000000	NaN	1.0	0.000000	NaN	0.000000	0.0	0.000000	0.000000
B22_M1	0.000000	0.000000	0.269231	0.5	NaN	0.357143	0.178571	NaN	0.0	1.000000	NaN	0.000000	0.0	0.000000	0.000000
B22_M2	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	1.0	NaN	NaN	NaN	NaN
B22_M3	0.000000	0.666667	0.000000	0.5	NaN	0.000000	0.000000	NaN	0.0	0.000000	NaN	1.000000	0.0	0.000000	0.000000
B22_M4	0.000000	0.200000	0.000000	0.0	NaN	0.000000	0.000000	NaN	0.0	0.000000	NaN	0.000000	1.0	0.000000	0.000000
B22_M5	0.000000	0.111111	0.000000	0.0	NaN	0.000000	0.111111	NaN	0.0	0.000000	NaN	0.000000	0.0	1.000000	0.000000
B22_M6	0.000000	0.000000	0.045455	0.0	NaN	0.071429	0.795455	NaN	0.0	0.000000	NaN	0.000000	0.0	0.000000	1.000000



	B5_M1	B5_M2	B5_M3	B5_M4	B6_M1	B6_M2	B6_M3	B6_M4	B6_M5	B6_M6
B5_M1	1.0	0.0	0.000000	NaN	0.0	1.000000	0.000000	0.00000	NaN	0.000000
B5_M2	0.0	1.0	0.000000	NaN	0.0	0.000000	0.000000	1.00000	NaN	0.000000
B5_M3	0.0	0.0	1.000000	NaN	0.0	0.000000	0.000000	0.00000	NaN	0.111111
B5_M4	NaN	NaN	NaN	1.0	NaN	NaN	NaN	NaN	NaN	NaN
B6_M1	0.0	0.0	0.000000	NaN	1.0	0.000000	0.000000	0.00000	NaN	0.000000
B6_M2	1.0	0.0	0.000000	NaN	0.0	1.000000	0.033333	0.00000	NaN	0.000000
B6_M3	0.0	0.0	0.000000	NaN	0.0	0.033333	1.000000	0.00000	NaN	0.000000
B6_M4	0.0	1.0	0.000000	NaN	0.0	0.000000	0.000000	1.00000	NaN	0.023810
B6_M5	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	1.0	NaN
B6_M6	0.0	0.0	0.111111	NaN	0.0	0.000000	0.000000	0.02381	NaN	1.000000

Figure 3. Conserved modules identified between the virulence interactomes of Salmonella (B5) and Vibrio cholerae (B6):

	module_a	bacterium_a	module_b	bacterium_b	comparison_type	comparison_status	terms_a_size	terms_b_size	intersection_size	union_size	overlap_coefficient
0	B5_M1	B5	B6_M2	B6	cross_species	comparable	10	31	10	31	1.000000
1	B5_M2	B5	B6_M4	B6	cross_species	comparable	1	42	1	42	1.000000
2	B5_M3	B5	B6_M6	B6	cross_species	comparable	9	67	1	75	0.111111

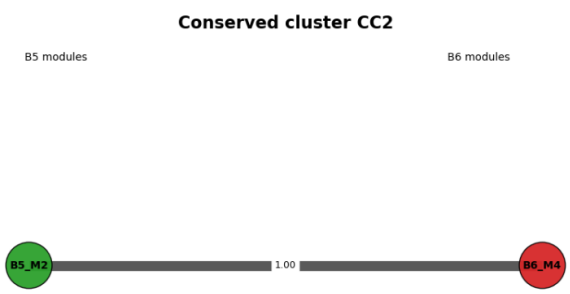


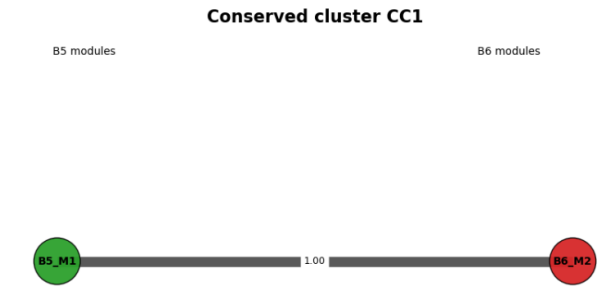
Cross-species conserved clusters on the clustered similarity graph

method	num_conserved_clusters	largest_cluster	avg_cluster_size	cross_bacteria_clusters	avg_num_bacteria_per_cluster	avg_internal_similarity	modularity
connected_components	3	2	2.0	3	2.0	0.7037	0.5485

Cross-species conserved-cluster table

	method	conserved_cluster_id	num_modules	num_bacteria	bacteria	modules	num_similarity_edges	avg_internal_similarity
0	connected_components	CC1	2	2	B5; B6	B5_M1; B6_M2	1	1.0000
1	connected_components	CC2	2	2	B5; B6	B5_M2; B6_M4	1	1.0000
2	connected_components	CC3	2	2	B5; B6	B5_M3; B6_M6	1	0.1111





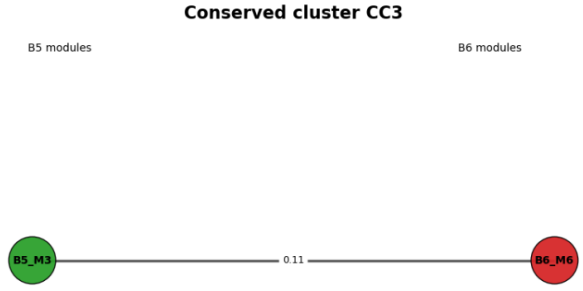
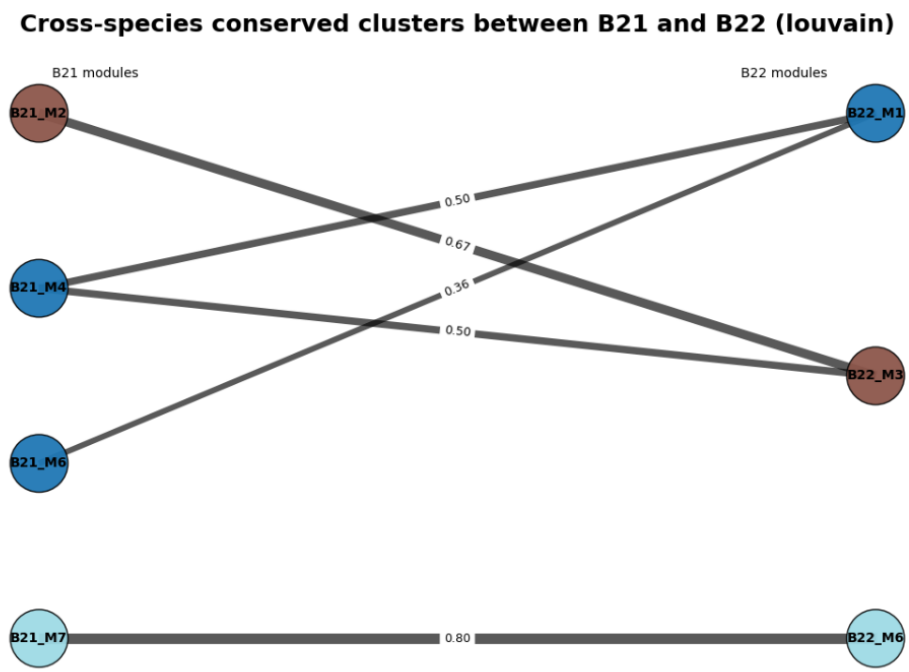


Figure 4. Conserved modules identified between the resistance interactomes of Bacillus subtilis (B21) and Erwinia (B22):

	module_a	bacterium_a	module_b	bacterium_b	comparison_type	comparison_status	terms_a_size	terms_b_size	intersection_size	union_size	overlap_coefficient
0	B21_M7	B21	B22_M6	B22	cross_species	comparable	56	44	35	65	0.795455
1	B21_M2	B21	B22_M3	B22	cross_species	comparable	34	6	4	36	0.666667
2	B21_M4	B21	B22_M1	B22	cross_species	comparable	4	56	2	58	0.500000
3	B21_M4	B21	B22_M3	B22	cross_species	comparable	4	6	2	8	0.500000
4	B21_M6	B21	B22_M1	B22	cross_species	comparable	14	56	5	65	0.357143

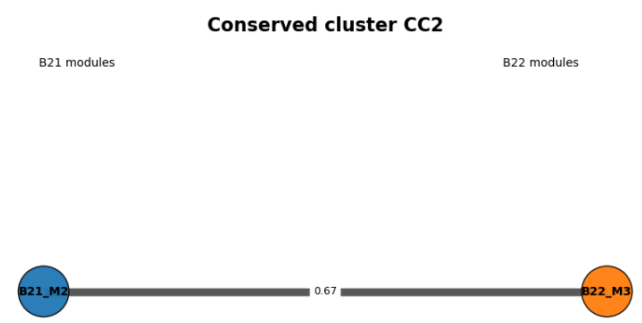


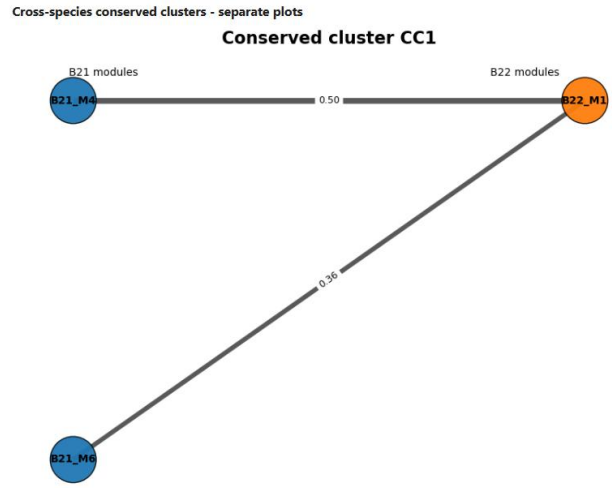
Cross-species conserved clusters on the clustered similarity graph

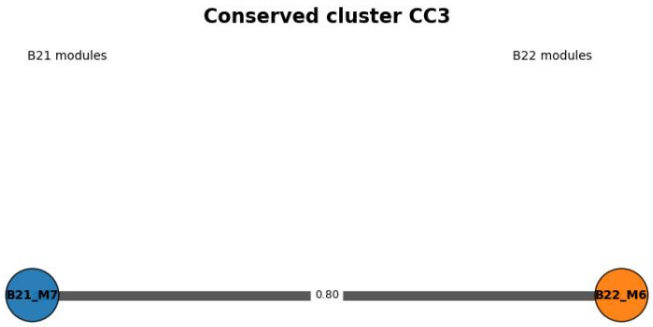
	method	num_conserved_clusters	largest_cluster	avg_cluster_size	cross_bacteria_clusters	avg_num_bacteria_per_cluster	avg_internal_similarity	modularity
0	louvain	3	3	2.33	3	2.0	0.6302	0.4831

Cross-species conserved-cluster table

	method	conserved_cluster_id	num_modules	num_bacteria	bacteria	modules	num_similarity_edges	avg_internal_similarity
0	louvain	CC1	3	2	B21; B22	B21_M4; B21_M6; B22_M1	2	0.4286
1	louvain	CC2	2	2	B21; B22	B21_M2; B22_M3	1	0.6667
2	louvain	CC3	2	2	B21; B22	B21_M7; B22_M6	1	0.7955







	module_a	bacterium_a	module_b	bacterium_b	comparison_type	comparison_status	terms_a_size	terms_b_size	intersection_size	union_size	overlap_coefficient
0	B21_M3	B21	B21_M4	B21	within_species	comparable	52	4	2	54	0.5

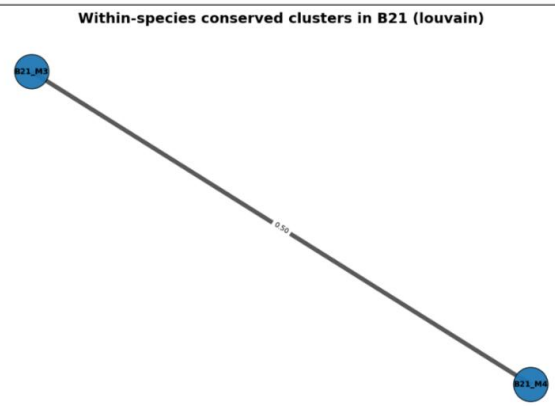


Figure 5. Higher-level module similarity network showing module families across more than two bacterial species (creative part):

Directory screening summary per target module

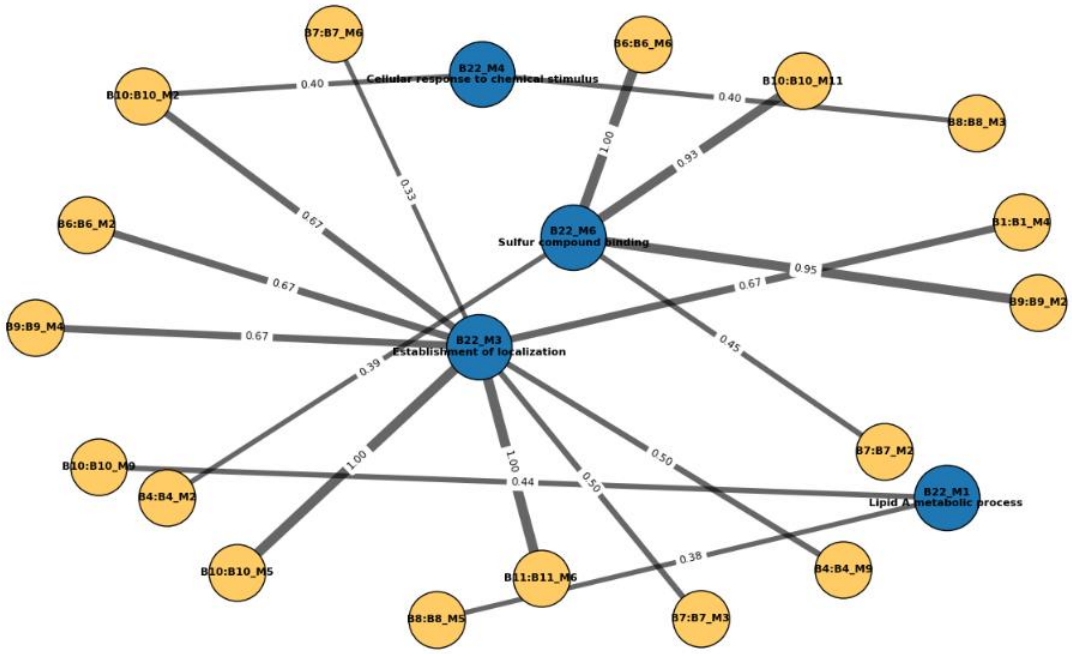
	target_module	target_best_term_description	target_best_term_category	target_num_significant_terms	support_inputs	support_bacteria	matched_inputs	matched_modules	matched_best_terms	avg_score	max_score	passes_directory_filter
0	B22_M3	Transmembrane transport	GO Process	6	7	7	1; 10; 11; 4; 6; 7; 9	B10_M2; B10_M5; B11_M6; B1_M4; B4_M9; B6_M2; B...	Efflux transmembrane transporter activity; Est...	0.6667	1.0000	True
1	B22_M6	Exopeptidase activity	GO Function	44	5	5	10; 4; 6; 7; 9	B10_M11; B4_M2; B6_M6; B7_M2; B9_M2	Cell division; Cell wall biogenesis; Exopeptid...	0.7462	1.0000	True
2	B22_M1	Oligosaccharide metabolic process	GO Process	56	2	2	10; 8	B10_M9; B8_M5	Antibiotic biosynthetic process; Pigment biosy...	0.4111	0.4375	True
3	B22_M4	Cellular response to antibiotic	GO Process	5	2	2	10; 8	B10_M2; B8_M3	Establishment of protein localization; Phospho...	0.4000	0.4000	True

Creative Extension Results

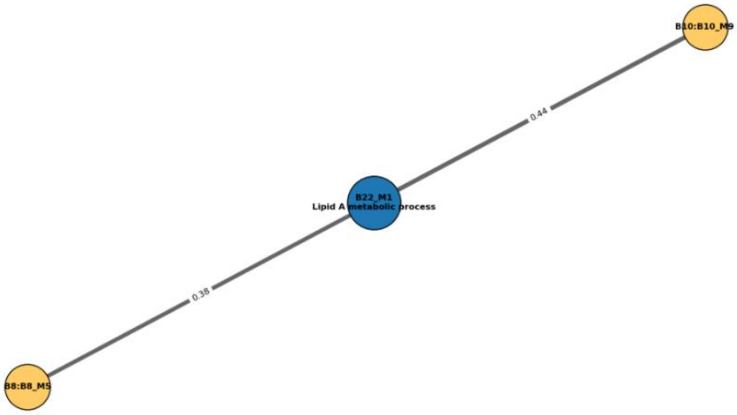
- Target folder: `./Input Resistance B21-B22`
- Target bacterium ID: `22`
- Screening folder: `./Input_Networks_Virulence`
- Similarity metric: `overlap_coefficient`
- Similarity threshold: `0.3`
- Minimum number of distinct screening inputs required: `2`

Directory screening graph for modules that passed the filter

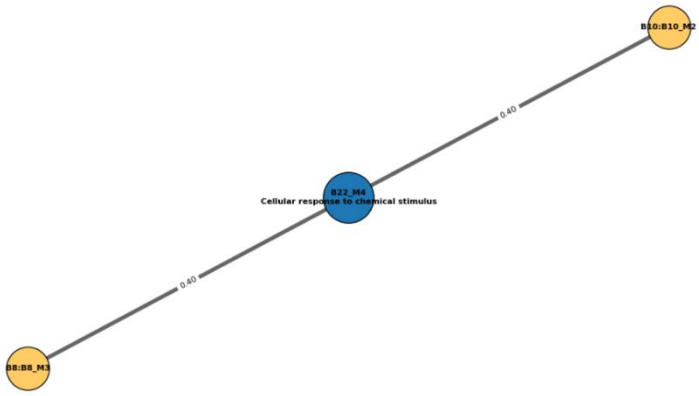
Directory screening graph (passed target modules only)

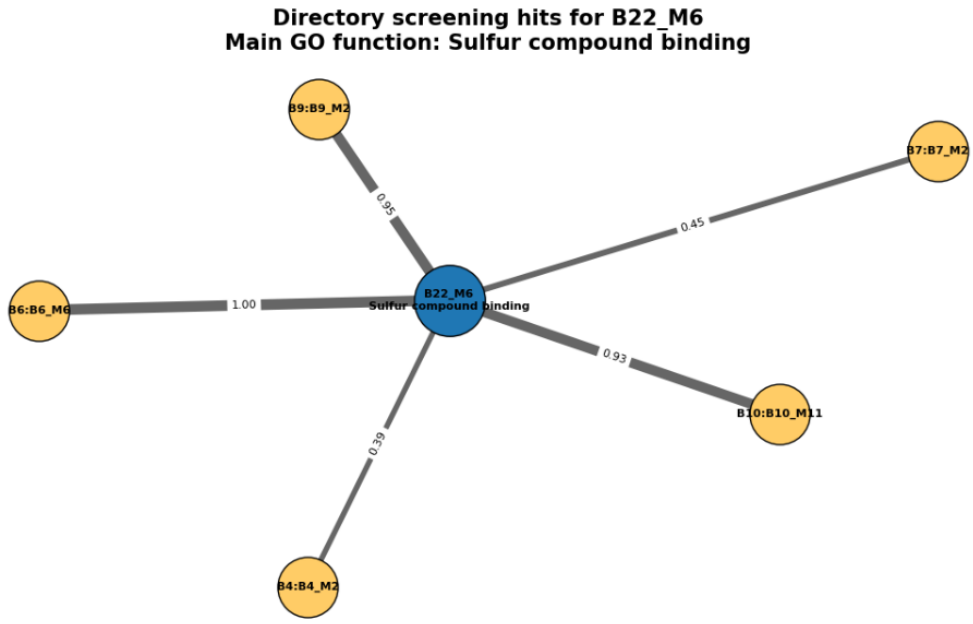


Directory screening hits for B22_M1
Main GO function: Lipid A metabolic process



Directory screening hits for B22_M4
Main GO function: Cellular response to chemical stimulus





Separate plots for each target module that passed the filter

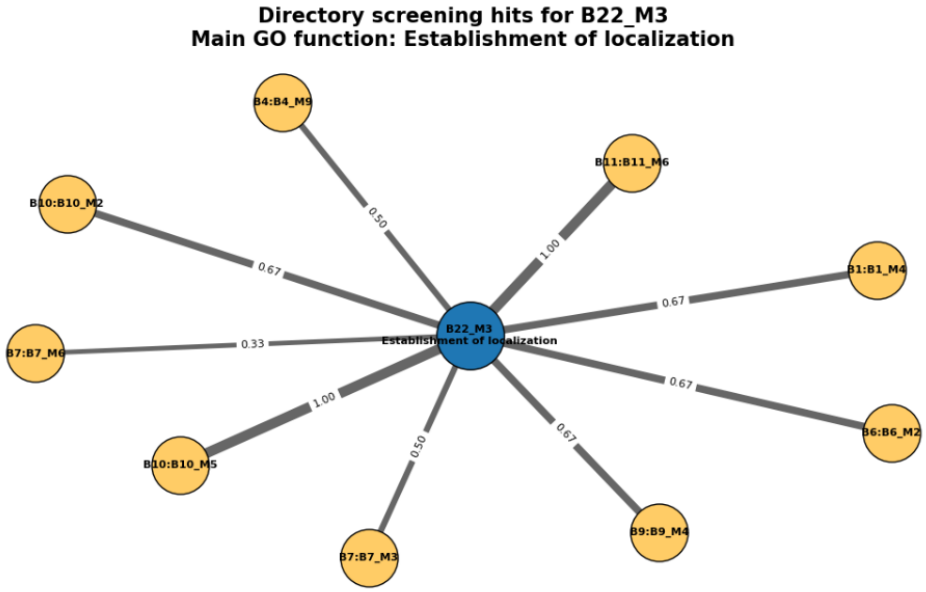


Table 1. Creative overview summary

	Target bacterium	Screened input count	Raw cross-species hits	Target modules in summary	Passed directory filter
0	B22	11	18	4	4

Table 2. Directory screening summary per target module

	target_module	target_best_term_description	target_best_term_category	target_num_significant_terms	support_inputs	support_bacteria	matched_inputs	matched_bacteria	matched_modules	matched_best_terms	avg_score	max_score	passes_directory_filter
0	B22_M3	Localization	GO Process	6	7	7	1; 10; 11; 4; 6; 7; 9	B1; B10; B11; B4; B6; B7; B9	B10_M2; B10_M5; B11_M6; B1_M4; B4_M9; B6_M2; B7_M3; B7_M6; B9_M4	Amide transport; Efflux transmembrane transporter activity; Establishment of localization; Intrinsic component of plasma membrane; Protein transmembrane transporter activity; Protein transport across the cell outer membrane; Response to antibiotic; Secretion by cell; Transporter activity	0.6667	1.0000	True
1	B22_M6	Catalytic activity, acting on a protein	GO Function	44	5	5	10; 4; 6; 7; 9	B10; B4; B6; B7; B9	B10_M11; B4_M2; B6_M6; B7_M2; B9_M2	Cell division; Cell wall macromolecule biosynthetic process; Cellular biosynthetic process; Serine-type D-Ala-D-Ala carboxypeptidase activity; Serine-type peptidase activity	0.7462	1.0000	True
2	B22_M1	Cellular polysaccharide biosynthetic process	GO Process	56	2	2	10; 8	B10; B8	B10_M9; B8_M5	Antibiotic biosynthetic process; Isoprenoid metabolic process	0.4111	0.4375	True
3	B22_M4	Cellular response to chemical stimulus	GO Process	5	2	2	10; 8	B10; B8	B10_M2; B8_M3	Amide transport; Phosphorelay signal transduction system	0.4000	0.4000	True

Table 3. Directory screening raw cross-species hits

	target_module	matched_module	screening_bacterium	screening_input_id	screening_score	intersection_size	union_size	target_best_term_description	target_best_term_category	target_num_significant_terms	matched_best_term_description	matched_best_term_category
0	B22_M3	B1_M4	B1	1	0.666667	2	7	Localization	GO Process	6	Transporter activity	GO Function
1	B22_M3	B10_M5	B10	10	1.000000	6	12	Localization	GO Process	6	Response to antibiotic	GO Process
2	B22_M6	B10_M11	B10	10	0.931818	41	74	Catalytic activity, acting on a protein	GO Function	44	Serine-type D-Ala-D-Ala carboxypeptidase activity	GO Function
3	B22_M3	B10_M2	B10	10	0.666667	4	51	Localization	GO Process	6	Amide transport	GO Process
4	B22_M1	B10_M9	B10	10	0.437500	7	65	Cellular polysaccharide biosynthetic process	GO Process	56	Antibiotic biosynthetic process	GO Process
5	B22_M4	B10_M2	B10	10	0.400000	2	52	Cellular response to chemical stimulus	GO Process	5	Amide transport	GO Process
6	B22_M3	B11_M6	B11	11	1.000000	6	8	Localization	GO Process	6	Intrinsic component of plasma membrane	GO Component
7	B22_M3	B4_M9	B4	4	0.500000	3	34	Localization	GO Process	6	Secretion by cell	GO Process
8	B22_M6	B4_M2	B4	4	0.390244	16	69	Catalytic activity, acting on a protein	GO Function	44	Cell division	GO Process
9	B22_M6	B6_M6	B6	6	1.000000	44	67	Catalytic activity, acting on a protein	GO Function	44	Serine-type peptidase activity	GO Function
10	B22_M3	B6_M2	B6	6	0.666667	4	33	Localization	GO Process	6	Protein transport across the cell outer membrane	GO Process
11	B22_M3	B7_M3	B7	7	0.500000	3	19	Localization	GO Process	6	Establishment of localization	GO Process
12	B22_M6	B7_M2	B7	7	0.454545	20	100	Catalytic activity, acting on a protein	GO Function	44	Cell wall macromolecule biosynthetic process	GO Process
13	B22_M3	B7_M6	B7	7	0.333333	2	24	Localization	GO Process	6	Protein transmembrane transporter activity	GO Function
14	B22_M4	B8_M3	B8	8	0.400000	2	144	Cellular response to chemical stimulus	GO Process	5	Phosphorelay signal transduction system	GO Process
15	B22_M1	B8_M5	B8	8	0.384615	10	72	Cellular polysaccharide biosynthetic process	GO Process	56	Isoprenoid metabolic process	GO Process
16	B22_M6	B9_M2	B9	9	0.954545	42	73	Catalytic activity, acting on a protein	GO Function	44	Cellular biosynthetic process	GO Process
17	B22_M3	B9_M4	B9	9	0.666667	2	7	Localization	GO Process	6	Efflux transmembrane transporter activity	GO Function

Table 1. Summary of modules detected in each bacterial network

	Bacterium	Number of modules	Largest module size	Smallest reported module size	Mean module size	Notes
0	B5	4	17	6	11.25	
1	B6	6	24	2	11.33	
2	B21	9	23	2	8.11	
3	B22	6	10	2	5.00	

Table 2. Summary of enriched biological terms associated with representative modules

	Bacterium	Module ID	Size	Top enriched term(s)	Functional interpretation
0	B5	B5_M1	17	Protein localization to extracellular region, Establishment of protein localization to extracellular region, Protein secretion	Protein localization to extracellular region
1	B5	B5_M3	8	Response to stimulus, Regulation of RNA metabolic process, Regulation of nucleic acid-templated transcription	Response to stimulus
2	B5	B5_M2	14	Extracellular region	Extracellular region
3	B6	B6_M6	2	Serine-type peptidase activity, Serine-type carboxypeptidase activity, Serine hydrolase activity	Serine-type peptidase activity
4	B6	B6_M4	4	Toxin activity, Modulation of process of another organism, Biological process involved in interspecies interaction between organisms	Toxin activity
5	B6	B6_M2	20	Protein transport across the cell outer membrane, Protein secretion by the type II secretion system, Secretion by cell	Protein transport across the cell outer membrane
6	B6	B6_M3	15	Negative regulation of cellular process, Cell-substrate adhesion, Cell adhesion involved in biofilm formation	Negative regulation of cellular process
7	B6	B6_M1	24	Cilium or flagellum-dependent cell motility, Archaeal or bacterial-type flagellum-dependent cell motility, Bacterial-type flagellum-dependent cell motility	Cilium or flagellum-dependent cell motility
8	B21	B21_M7	3	Serine-type D-Ala-D-Ala carboxypeptidase activity, Carboxypeptidase activity, Serine-type exopeptidase activity	Serine-type D-Ala-D-Ala carboxypeptidase activity
9	B21	B21_M3	10	Gene expression, Cellular nitrogen compound biosynthetic process, Response to antibiotic	Gene expression
10	B21	B21_M1	23	Regulation of cellular process, Regulation of sporulation, Regulation of biological process	Regulation of cellular process
11	B21	B21_M2	14	Amide transport, Establishment of protein localization, Cellular macromolecule localization	Amide transport
12	B21	B21_M6	4	Antibiotic biosynthetic process, Antibiotic metabolic process, Ligase activity	Antibiotic biosynthetic process
13	B21	B21_M9	2	ATP-dependent activity, Primary active transmembrane transporter activity, ABC-type transporter activity	ATP-dependent activity
14	B21	B21_M4	8	Response to antibiotic, Response to chemical, Transporter activity	Response to antibiotic
15	B22	B22_M1	10	Cellular polysaccharide biosynthetic process, Lipid A biosynthetic process, Phospholipid metabolic process	Cellular polysaccharide biosynthetic process
16	B22	B22_M6	2	Catalytic activity, acting on a protein, Penicillin binding, Carboxylic acid binding	Catalytic activity, acting on a protein
17	B22	B22_M5	2	Cation transmembrane transport, Cation transmembrane transporter activity, Spermidine transport	Cation transmembrane transport
18	B22	B22_M3	7	Localization, Establishment of localization, Transporter activity	Localization
19	B22	B22_M4	2	Cellular response to chemical stimulus, Cellular response to antibiotic, Cellular response to stimulus	Cellular response to chemical stimulus

Table 3. Summary of conserved modules detected in B5 versus B6 and in B21 versus B22

	Pair	Bacterium A	Module A	Bacterium B	Module B	Similarity (overlap_coefficient)	Shared GO terms	Union GO terms
0	B5 vs B6	B5	B5_M1	B6	B6_M2	1.000000	10	31
1	B5 vs B6	B5	B5_M2	B6	B6_M4	1.000000	1	42
2	B5 vs B6	B5	B5_M3	B6	B6_M6	0.111111	1	75
3	B21 vs B22	B21	B21_M7	B22	B22_M6	0.795455	35	65
4	B21 vs B22	B21	B21_M2	B22	B22_M3	0.666667	4	36
5	B21 vs B22	B21	B21_M4	B22	B22_M1	0.500000	2	58
6	B21 vs B22	B21	B21_M4	B22	B22_M3	0.500000	2	8
7	B21 vs B22	B21	B21_M3	B21	B21_M4	0.500000	2	54
8	B21 vs B22	B21	B21_M6	B22	B22_M1	0.357143	5	65

Table 4. Summary of identified module families, species composition and dominant functions

	Pair	Module family	Species composition	Modules	Dominant biological functions
0	B5 vs B6	CC1	B5; B6	B5_M1; B6_M2	Protein localization to extracellular region; Protein transport across the cell outer membrane
1	B5 vs B6	CC2	B5; B6	B5_M2; B6_M4	Extracellular region; Toxin activity
2	B5 vs B6	CC3	B5; B6	B5_M3; B6_M6	Response to stimulus; Serine-type peptidase activity
3	B21 vs B22	CC1	B21; B22	B21_M4; B21_M6; B22_M1	Response to antibiotic; Antibiotic biosynthetic process; Cellular polysaccharide biosynthetic process
4	B21 vs B22	CC2	B21; B22	B21_M2; B22_M3	Amide transport; Localization
5	B21 vs B22	CC3	B21; B22	B21_M7; B22_M6	Serine-type D-Ala-D-Ala carboxypeptidase activity; Catalytic activity, acting on a protein

12. Bibliography

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- Sharan, R., et al. (2005). Conserved patterns of protein interaction in multiple species. Proceedings of the National Academy of Sciences (PNAS).